

✓ SAA - ACTIVIDAD 3.19 - ÁLVARO FERNÁNDEZ BECERRA

Entrena un clasificador basado en AdaBoostClassifier de Scikit-Learn para el dataset de MNIST.

Realiza validación simple con un 80% de datos de entrenamiento y el resto para validación.

Obtén la tasa de aciertos (accuracy score) para el dataset de validación.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.datasets import fetch_openml
import random

mnist = fetch_openml('mnist_784')
x = mnist.data
y = mnist.target
# Mostrar varios dígitos al azar
```

✓ Split

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=33)
```

✓ Adaboost:

```
clf = AdaBoostClassifier(random_state = 33)
clf.fit(x_train, y_train)

# Validación
y_pred = clf.predict(x_test)

print("accuracy score: ", accuracy_score(y_test, y_pred))
```

```
accuracy score: 0.6469285714285714
```

- Al final se obtiene un `0.6469285714285714` de accuracy